

Agile:

◆ Agile Important Questions

1. What is Agile?

- Agile is a **software development methodology** that emphasizes iterative development, collaboration, flexibility, and delivering value quickly.
 - Key idea: Break projects into **small increments (sprints)** instead of long, rigid plans.
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2. Agile vs Waterfall

Aspect	Agile	Waterfall
Approach	Iterative & incremental	Sequential
Flexibility	High, can adapt to changes	Low, changes expensive
Delivery	Frequent releases	At end of project
Testing	Continuous	After development
Team	Cross-functional, collaborative	Specialized roles

3. Key Agile Principles

1. Customer satisfaction through continuous delivery.
2. Welcome changing requirements.
3. Frequent delivery of working software.
4. Collaboration between business and developers.
5. Sustainable development pace.
6. Simplicity and technical excellence.

7. Self-organizing teams.
 8. Regular reflection and improvement.
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4. Scrum vs Kanban

- **Scrum:**
 - Iterative sprints (2–4 weeks), fixed roles (Scrum Master, Product Owner, Dev Team), ceremonies (Daily Standup, Sprint Planning, Retrospective).
 - **Kanban:**
 - Continuous flow, no fixed sprints, focuses on **visualizing work** (Kanban board) and limiting WIP (Work In Progress).
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5. Roles in Scrum

1. **Product Owner:** Defines requirements, prioritizes backlog.
 2. **Scrum Master:** Facilitates process, removes blockers.
 3. **Development Team:** Builds the product, self-organizing.
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6. What is a Sprint?

- A time-boxed iteration (usually 2–4 weeks) where the team delivers a **potentially shippable product increment**.
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7. What is a Product Backlog vs Sprint Backlog?

- **Product Backlog:** Ordered list of all features, enhancements, and bug fixes for the product.
 - **Sprint Backlog:** Subset of the product backlog that the team commits to complete in the current sprint.
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8. What is a User Story?

- Short, simple description of functionality from the **user's perspective**.
- Format: *"As a [user], I want [feature] so that [benefit]."*

- Example: *"As a shopper, I want to filter products by price so that I can find affordable items."*
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9. What is Definition of Done (DoD)?

- Criteria that must be met for a user story to be considered **complete**.
 - Example: Code written, unit tests passed, code reviewed, deployed to staging, documentation updated.
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10. Agile Metrics / KPIs

- **Velocity:** Story points completed per sprint.
 - **Burndown Chart:** Remaining work vs time.
 - **Cumulative Flow Diagram:** Work in different states over time.
 - **Lead Time / Cycle Time:** Time to complete a task from start to finish.
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11. Difference between Story Points vs Hours

- **Story Points:** Relative measure of effort/complexity.
 - **Hours:** Absolute time estimate.
 - Agile prefers story points for planning velocity and team capacity.
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12. How to handle changing requirements mid-sprint?

- Ideally, **avoid scope changes mid-sprint**.
 - If urgent, team and PO negotiate **priority swap or sprint re-planning**.
 - Agile welcomes changes but balances with team commitment.
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13. What is a Daily Standup?

- 15-minute daily meeting where team answers:
 1. What did I do yesterday?
 2. What will I do today?

3. Any blockers?

14. What is a Retrospective?

- Meeting at end of sprint to discuss:
 - What went well
 - What didn't
 - Actions to improve next sprint
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15. How do you measure Agile success?

- Delivery of working software frequently.
 - Team velocity consistency.
 - Customer satisfaction and feedback.
 - Reduction in defect rate.
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16. Agile Artifacts

- **Product Backlog** – all features/bugs prioritized
 - **Sprint Backlog** – tasks for current sprint
 - **Increment** – working product delivered after each sprint
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17. How do you handle blockers?

- Raise in **Daily Standup**.
 - Scrum Master facilitates resolution.
 - Escalate if necessary.
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18. Agile Ceremonies

- Sprint Planning, Daily Standup, Sprint Review, Retrospective.
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19. Agile Testing Practices

- Continuous integration with automated tests
 - Test-driven development (TDD)
 - Exploratory testing during sprint
 - Regression testing before release
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20. Difference between Epic, Story, Task, Sub-task

Term	Description
Epic	Large feature spanning multiple sprints
Story	User-focused requirement
Task	Technical work needed for story
Sub-task	Smaller chunk of task, assigned to individual

21. Scaling Agile

- **SAFe / LeSS / Nexus:** Frameworks to scale Agile across multiple teams.
 - Focuses on coordination, shared backlog, and cross-team alignment.
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22. How do Agile teams handle estimation?

- Use **Planning Poker / Story Points**
 - Consider complexity, effort, risk, dependencies
 - Team consensus rather than individual estimate
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23. Difference between Incremental vs Iterative development

- **Incremental:** Build product piece by piece (functionality added).
 - **Iterative:** Improve/refine product through repeated cycles based on feedback.
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24. What is Continuous Delivery in Agile?

- Automated build, test, and deployment pipeline ensuring code is **always ready for production**.

- Agile + CI/CD → faster feedback and delivery.
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25. How to handle distributed Agile teams?

- Use **video calls, chat, shared tools (Jira, Confluence)**
 - Clear documentation, overlapping hours for standups
 - Agile ceremonies adapted to remote context
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